SUNNY GEORGE CHANDY

DIRECTOR, AI/ML ENGINEERING | ENTERPRISE AI & ENGINEERING LEADERSHIP

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EXECUTIVE PROFILE

Engineering and AI leader with 15+ years of experience building enterprise software, modernizing mission-critical platforms, and delivering production AI in regulated healthcare and financial services. Lead a 40-member global engineering organization across AI, software development, quality engineering, SRE, and operations. Combine hands-on agentic AI architecture with engineering strategy, dependable delivery, responsible AI, and measurable business results.

LEADERSHIP FOCUS

Global Engineering Leadership • AI Strategy & Architecture • Agentic AI & LLM Evaluation • Software Delivery • Cloud & SRE • DevOps & Observability • Responsible AI

PROFESSIONAL EXPERIENCE

Director, AI/ML Engineering

Optum Financial Services (Optum) | Minneapolis, MN | Jul 2024 – Present

- **Engineering leadership.** Lead a 40-member global organization spanning AI/ML engineering, software development, quality engineering, site reliability engineering, and production operations; set technical direction, delivery standards, and cross-functional priorities.
- **Enterprise AI at scale.** Delivered conversational AI supporting 220K+ daily interactions and 3M+ monthly conversations, achieving 75% containment, 94% CSAT, 99.99% uptime, and $8M+ in reported annual operational savings.
- **Agentic AI delivery.** Led development of an omnichannel multi-agent platform using Google ADK, MCP, RAG, memory, and model routing; integration with Employer IVR improved containment from 50% to 65%.
- **AI evaluation & performance.** Established automated LLM evaluation using golden datasets, LLM-as-judge, rules, and human review; developed an agent-driven k6 framework to test quality and performance at 100–1,000 concurrent users.
- **SRE & platform engineering.** Lead the team responsible for cloud infrastructure provisioning, CI/CD pipelines, release automation, application observability, reliability, and operational readiness for production AI systems across Azure and GCP.
- **AI analytics & observability.** Delivered conversation analytics with resolution detection and repeat-caller insights, reducing investigation time by 90% and improving operational visibility.
- **Strategy, governance & adoption.** Partner with Product, Operations, Security, and business leadership on AI roadmaps, production readiness, and responsible AI controls; building a Forward Deployed Engineering practice to accelerate use-case delivery.

PROFESSIONAL EXPERIENCE — CONTINUED

Senior Engineering Manager – Software Engineering & Operations

Optum Pay (Optum) | Minneapolis, MN | Apr 2017 – Jun 2024

- **Engineering organization.** Led software developers, quality engineers, SRE, DevOps, and IT operations supporting healthcare payment systems processing $300B+ annually for 2M+ healthcare professionals.
- **Reliability & observability.** Sustained 99.995% platform uptime; implemented end-to-end application observability across services and infrastructure to improve monitoring, troubleshooting, and incident response.
- **Cloud modernization & security.** Led Azure application and infrastructure modernization with zero SLA breaches and reduced high/critical vulnerabilities by more than 99% (100K+ to fewer than 100).
- **Critical payment infrastructure.** Scaled infrastructure supporting $135.2B in HHS COVID-19 Provider Relief Fund payments to more than 421K healthcare providers.
- **Crisis analytics.** Delivered reporting dashboards tracking $6.5B in Temporary Funding Assistance during the Change Healthcare outage, enabling operational and payment visibility.
- **Resilience engineering.** Introduced Optum Financials' first chaos engineering initiative using Azure Chaos Studio to identify failure modes and strengthen distributed-system resilience.

IT Analyst – Software Engineering

Tata Consultancy Services (Client: Target) | Minneapolis, MN | Mar 2011 – Apr 2017

- **Software development.** Progressed from developer to technical lead, building retail and pharmacy applications using C#, .NET, JavaScript, and SQL Server; led cross-shore teams of 10+.
- **Infrastructure automation.** Developed provisioning, monitoring, and lifecycle automation that reduced manual work by approximately 60%.
- **Pharmacy systems.** Supported Target RedRx prescription fulfillment and claims adjudication through application development, SQL tuning, and production support.

TECHNICAL EXPERTISE

AI & Agentic Systems	LLMs, Generative AI, RAG, LangChain, Semantic Kernel, Google ADK, MCP, Dialogflow CX
Languages & Frameworks	Python, Java, C#, JavaScript, Bash, J2EE / Java EE, ASP.NET, Node.js
Cloud & Infrastructure	Azure, GCP, Docker, Kubernetes, Red Hat Enterprise Linux, CentOS
DevOps & Reliability	GitHub Actions, GitOps, Jenkins, JFrog Artifactory, CI/CD, SRE, performance testing
Observability	Datadog, Dynatrace, Grafana, Splunk, Fluentd
Databases & App Servers	PostgreSQL, IBM Db2, SQL Server, Oracle, Redis, Microsoft IIS, IBM WebSphere, Apache Tomcat

EDUCATION & PROFESSIONAL DEVELOPMENT

Boston University — M.S., Computer Science & Artificial Intelligence (in progress; expected Jan 2028)
Johns Hopkins University — Certificate, AI in Healthcare (Aug 2026)
The University of Texas at Austin — Post Graduate Program, Generative AI (2025–2026)
Karunya Institute of Technology and Sciences — B.E., Computer Science & Engineering (2010)